

4IR Technologies Individual Assignment

Mock AI-Assisted Model Submission Draft

Use note: This is a mock AI-assisted draft for study, rewriting and declaration. Before submission, rewrite it in your own voice, verify all sources, complete the official AI declaration form, and add your campus and student details.

Faculty	Information Technology
Module Code	ITTNA3-22
Module Name	4IR Technologies
Assessment	Individual Assignment
Student	[Insert your name and student number]
Campus	[Insert campus name]

Introduction

The Fourth Industrial Revolution (4IR) refers to the integration of digital, physical, biological and cyber technologies into everyday economic and social systems. Technologies such as artificial intelligence, robotics, the Internet of Things, cloud computing, biotechnology and digital platforms are changing how people learn, work, communicate, produce goods and access services. In the education sector, 4IR is especially important because universities are expected not only to teach new technologies, but also to use them for research, innovation and community development. However, the benefits of 4IR are not automatic. In Africa and South Africa, 4IR can create new opportunities only if inequalities in access, skills, infrastructure and participation are addressed. This assignment argues that 4IR should be approached as an inclusive development tool rather than as technology adoption for its own sake.

1.1 Inequalities Affecting Innovation And Research In African Universities

The first major inequality is digital access inequality. Students and researchers do not all have equal access to suitable devices, affordable data, stable electricity, reliable internet connections and digital learning platforms. This directly affects innovation because modern research often depends on online journals, digital libraries, cloud storage, video collaboration, data collection platforms and specialised software. A student with only occasional mobile data access cannot participate in online research work in the same way as a student with a laptop and stable fibre connection. In South Africa, Statistics South Africa (2025) reports that 85.6% of households had some form of internet access, but only 20.6% had fixed internet access at home, while 78.9% relied on mobile access. This shows that access exists, but the quality and stability of access remain unequal. For African universities, this can slow collaborative research, reduce student participation in online learning, and prevent researchers from using data-intensive 4IR tools. Practical strategies include university-negotiated data bundles, device-loan programmes, extended campus Wi-Fi zones, zero-rated academic platforms, open computer labs and community digital access centres linked to universities.

The second inequality is research funding and infrastructure inequality. Many African institutions do not have the same access to advanced laboratories, modern equipment, licensed software, cloud computing resources, research grants and technology-transfer offices as better-resourced universities elsewhere. This means that researchers may identify important local problems but lack the tools to prototype, test, publish or commercialise their solutions. For example, a university may have researchers who understand local agricultural, health or energy challenges, but without laboratories, sensors, datasets or funding, those ideas may remain theoretical. UNESCO (2024) notes that underinvestment in higher education research and development limits Africa's innovation capacity, especially in fields such as health, agriculture and technology. This inequality weakens the continent's ability to produce local solutions for local problems. Strategies to address it include shared regional research hubs, government-funded innovation grants, industry-sponsored laboratories, open innovation challenges, cloud-credit partnerships and stronger collaboration between universities, businesses and public-sector institutions.

The third inequality is skills and curriculum inequality. 4IR innovation requires more than access to technology; it also requires people who know how to use technology critically and creatively. Some institutions update their curricula quickly to include artificial intelligence, robotics, data analytics, cybersecurity, cloud platforms and the Internet of Things, while others remain focused on older theoretical content. This creates uneven graduate readiness and limits the ability of students to participate in 4IR research. A student who has never worked with data tools or practical digital projects may struggle to contribute to research that uses machine learning, sensors or automation. The World Economic Forum (2025) reports that more than 60% of businesses operating in South Africa identify skills gaps as a major barrier to business transformation. This suggests that universities need to align

learning with industry and community needs. Practical strategies include lecturer upskilling, industry co-designed modules, short certification programmes, hackathons, work-integrated learning and project-based assessments where students solve real problems using 4IR tools.

The fourth inequality is gender and socio-economic inequality. Women, poorer students and historically disadvantaged communities often face barriers to STEM subjects, funding, professional networks, mentoring and safe innovation spaces. This affects research because innovation is stronger when different groups contribute different experiences and problem-solving perspectives. If disadvantaged groups are excluded, universities lose potential innovators who understand community problems from lived experience. For example, a young woman from an under-resourced township may have valuable insight into digital education, safety, transport or small-business challenges, but without scholarships, mentorship and access to STEM pathways, her ideas may never reach a research or entrepreneurial stage. UNESCO (2024) highlights persistent gender disparities in STEM fields and the importance of women's participation in higher education for development. Strategies should therefore include targeted scholarships, women-in-STEM mentorship, outreach to under-resourced schools, inclusive innovation competitions, entrepreneurship support and partnerships that deliberately include students from lower-income backgrounds.

Overall, these inequalities are connected rather than separate. Weak digital access reduces skills development, poor funding limits research infrastructure, outdated curricula limit innovation capacity, and social exclusion reduces the diversity of ideas. African universities therefore need integrated strategies that address access, funding, skills and inclusion together.

1.2 Digital Technology Project For An Urban South African Community

If I were selected as one of the 21 African leaders in the AfricaInLead programme, I would choose digital technology as the foundation for a community project. The proposed project is an Urban Digital Skills and Micro-Innovation Hub. The socio-economic challenge it addresses is urban youth unemployment, digital exclusion and the lack of practical work experience among young people. South Africa's unemployment context makes this project relevant. Statistics South Africa (2026) reports that the national unemployment rate was 32.7% in Q1 2026, while youth aged 15-24 had an unemployment rate of 60.9% and those aged 25-34 had a rate of 40.6%. Many young people may have access to smartphones, but they do not necessarily have employable digital skills, industry networks or portfolios of completed work. The project would focus on turning basic digital access into practical digital capability.

The hub would offer short, practical training programmes in coding basics, digital marketing, data literacy, cybersecurity awareness, cloud collaboration and entrepreneurship. Learners would not only complete theory-based lessons; they would work in teams on real community projects. For example, one group could build a basic website for a local small business, another could create a booking form for

a hair salon or tutoring service, and another could build a simple stock tracker for a spaza shop. The hub would use mobile-first learning because mobile internet is more common than fixed home internet in South Africa (Statistics South Africa, 2025). AI-assisted tutoring tools could support practice, explanations and coding help, but mentors would teach learners how to use these tools ethically, critically and responsibly. This is important because 4IR education should not train students to depend blindly on technology; it should train them to use technology to solve real problems.

The expected impact would be both individual and community-based. For young people, the hub would provide practical skills, confidence, career exposure and a portfolio that can be shown to employers or clients. This is stronger than a certificate alone because employers increasingly want evidence of applied ability. For local businesses, the hub would provide affordable digital support, such as websites, social media content, simple data systems and online booking tools. This can help small businesses become more visible, organised and competitive. For the wider community, the project would create a local ecosystem where young people are not only job seekers but also digital problem-solvers. The project's impact could be measured through course completion rates, number of community projects delivered, internships secured, income-generating freelance projects, small ventures created and learners who return as mentors.

The mentorship model would be central to sustainability. The first strategy would be a train-the-trainer model where the strongest graduates become peer mentors for the next cohort. This allows knowledge to remain in the community instead of depending only on outside experts. The second strategy would be partnerships with universities, local businesses, NGOs and technology professionals who can provide guest lectures, project briefs, equipment and career advice. The third strategy would be monthly showcase days where learners present their projects to community members, potential employers and small-business clients. Each learner would ideally have access to three types of mentorship: a technical mentor, a career mentor and a community project mentor. Funding could come from sponsorships, university partnerships, small service fees from businesses that can afford to pay, and local government or NGO support. In this way, the project would not be a once-off training event, but a sustainable pipeline for digital leadership and community innovation.

1.3 Comparing Japan And South Africa On Robotics

Japan and South Africa both face labour-market pressure, but the nature of the pressure is very different. In Japan, the issue discussed in the scenario is labour shortage, especially in logistics, where new working-time regulations created concerns about delivery capacity. The International Federation of Robotics (2023) explains that robots can support loading, unloading and warehouse tasks, helping to close productivity gaps and reduce pressure on workers. In South Africa, the challenge is almost the opposite. The country has high unemployment, especially among young people (Statistics South Africa, 2026). This means that robotics cannot simply be justified as a replacement for missing workers. Instead,

South Africa must use robotics in a way that improves productivity, safety and competitiveness while also creating new pathways into skilled work.

In Japan, robotics is easier to justify because the economy faces a shortage of available workers in certain sectors. The IFR (2023) notes that heavy truck drivers in Japan worked long annual hours and that robots can reduce time spent on loading and unloading. Robots can therefore support overworked employees and reduce the burden of dirty, dull and dangerous work. In logistics, mobile robots can move goods through warehouses, while industrial robots can support manufacturing and service robots can assist in care and hospitality environments. The main benefit is that robots help maintain economic activity when there are not enough workers or when working hours need to be reduced for health and safety reasons. In this context, automation protects productivity and can improve working conditions.

South Africa's situation is more complex because high unemployment creates fear that robots will replace workers. This concern is valid if automation is introduced only to cut labour costs. However, South Africa's problem is not only the number of jobs; it is also the mismatch between available skills and the needs of a changing economy. The World Economic Forum (2025) identifies skills gaps as a major barrier to business transformation in South Africa and notes that roles linked to AI, machine learning, robotics and digital transformation are expected to grow. This suggests that the country should not reject robotics, but should connect robotics adoption to training, technical education and job transition strategies. The key difference is that Japan mainly uses robots to fill labour gaps, while South Africa must use robots to create new value and develop human capability.

There are also similarities between the two countries. Both need productivity improvements, safer workplaces and more efficient service delivery. Both can use robotics to handle repetitive, dangerous or physically demanding tasks. The difference lies in implementation. In Japan, automation responds to a shortage of workers and an ageing population. In South Africa, automation must respond to unemployment, inequality and low productivity at the same time. Therefore, South Africa's policy response should be automation plus reskilling, local manufacturing, technical support services and worker consultation. If robots are imported and installed without developing local skills, the benefits will be limited. If robotics is linked to training technicians, programmers, operators and maintenance workers, it can support industrial upgrading and employment creation.

Despite high unemployment, robotics can benefit South Africa in several practical ways. In mining, robots and remote-controlled systems can reduce workers' exposure to dangerous underground environments. This is important because mining remains a significant sector, but it carries serious safety risks. In logistics and warehousing, robots can improve stock movement, reduce delays and support e-commerce growth, while creating demand for technicians who install, maintain and troubleshoot automated systems. In agriculture, drones, sensors and automated irrigation systems can improve crop monitoring, water use and food production. In healthcare, robots can assist with repetitive delivery tasks in hospitals, allowing nurses and healthcare workers to focus more on patient care. In

manufacturing, robotics can improve product quality, reduce waste and help local firms compete with international producers. These examples show that robotics can support human work rather than automatically eliminate it.

The critical condition is that South Africa should adopt robotics as worker augmentation, not simple worker replacement. This requires deliberate policy. Companies that automate should be encouraged or required to invest in reskilling affected workers. TVET colleges, universities and private training providers should offer practical robotics, mechatronics, data and maintenance programmes. Government can support local robotics suppliers and small businesses that provide installation, repair and support services. Workers should also be included in automation planning so that technology is introduced transparently and fairly. Japan shows that robots can solve labour shortages, but South Africa must adapt robotics to a different reality. The country can benefit from robotics if it treats automation as part of inclusive industrial development rather than as a shortcut to reduce labour costs.

1.4 Sharing Economy And Disposable Income

The sharing economy refers to the use of digital platforms to connect people who own underutilised assets with people who need temporary access to those assets. These assets can include spare rooms, cars, tools, bicycles, books, Wi-Fi, solar power or even skills. The important point is that ownership does not always transfer. Instead, people rent, lend, share or exchange access for a limited time. The assignment scenario uses Airbnb as an example because it is one of the largest accommodation platforms in the world even though it does not own the homes listed on the platform. Filippas, Horton and Zeckhauser (2020) explain that internet-based sharing-economy markets allow consumer-owners to rent durable goods to non-owners. This model can increase disposable income because it creates new earning opportunities for owners and lowers costs for users.

For asset owners, the sharing economy increases disposable income by turning idle capacity into extra earnings. A spare room, unused parking space, car, bicycle or power tool may sit unused for many hours of the day. Even when unused, these assets may still cost money to maintain. By sharing them through a platform or community system, households can generate income from resources they already own. For example, a family with a spare room could rent it to a student, tourist or temporary worker. A car owner could use the vehicle for e-hailing, delivery services or community car-sharing. A person with power tools could rent them to someone doing a once-off repair or small construction job. This income can help households cover food, electricity, transport, school fees, debt payments or savings. In this way, the sharing economy can improve disposable income without requiring households to buy new assets first.

For users, the sharing economy increases disposable income by reducing the need for expensive purchases. A person who needs a drill for one afternoon does not need to buy a full toolkit. A student who needs transport occasionally may use a shared bicycle or e-hailing service instead of owning a car. A

small business owner may rent equipment for a short project instead of taking on debt to buy it. This matters because high upfront costs can prevent people from starting income-generating activities. In the assignment scenario, the example of tools for a handyman job is useful because access to tools can be a barrier to entry. If tools can be rented affordably, someone can complete a paid job and earn income without first owning all the equipment. This means the sharing economy can support micro-enterprise, informal work and household resilience.

The sharing economy can also strengthen communities when it is applied beyond commercial platforms. A community tool library could allow households to borrow equipment for repairs. A bicycle-sharing system could support learners, workers and delivery riders. Skills-sharing platforms could allow people to exchange tutoring, coding, repairs, farming advice, design services or language support. Shared Wi-Fi or solar charging stations could reduce costs for households that cannot afford individual services. Van Embden (2018) argues that underutilised assets can be shared on a needs basis and monetised for owners, creating lower marginal costs and broader access. In a South African context, this idea is powerful because many households face financial pressure and cannot afford to own everything they need. Sharing can therefore increase both income and access.

However, a balanced answer must recognise that the sharing economy does not benefit everyone equally. People who already own valuable assets are often in a better position to earn from sharing. People with no spare room, no car, no tools and limited internet access may benefit mainly as users, not as asset owners. There are also risks such as platform fees, damage to property, safety concerns, insurance, tax obligations, unfair rating systems and digital exclusion. For the sharing economy to increase disposable income fairly, communities need trust, affordable internet, consumer protection and fair platform rules. Government and platform providers should also ensure that workers and asset owners are not exploited. Therefore, sharing underutilised assets can increase disposable income, but the model works best when access, safety, trust and regulation are taken seriously.

Conclusion

The 4IR creates major opportunities for African universities, South African communities, businesses and households, but those opportunities depend on inclusive implementation. African universities must address digital access, research infrastructure, curriculum relevance and social inclusion if they want to produce meaningful innovation. A digital skills and micro-innovation hub can help urban youth move from basic technology use to practical digital capability. Robotics can benefit South Africa if it improves safety and productivity while creating skilled pathways into technical work. The sharing economy can increase disposable income by helping households earn from underutilised assets and reduce the cost of access. Overall, 4IR should be understood as a human-centred development opportunity, not only a technological trend.

An important final point is that these technologies should be evaluated by the social value they create, not only by their technical sophistication. In each question, the strongest outcome depends on linking technology with inclusion: universities need access and diverse participation; communities need digital skills and mentorship; businesses need robotics with reskilling; and households need sharing platforms that are safe and fair. This makes 4IR a development strategy rather than a narrow technology trend.

Reference List

Evans, S. 2023. Robots could help Japan solve 2024 labor shortages. Available at:

<https://www.iotworldtoday.com/robotics/robots-could-help-japan-solve-2024-labor-shortages>
[Accessed 4 June 2026].

Filippas, A., Horton, J.J. and Zeckhauser, R.J. 2020. Owning, Using, and Renting: Some Simple Economics of the Sharing Economy. *Management Science*, 66(9), pp.4152-4172. Available at:

<https://www.hks.harvard.edu/publications/owning-using-and-renting-some-simple-economics-sharing-economy-0> [Accessed 4 June 2026].

International Federation of Robotics. 2023. Robots help to solve Japan's 2024 problem. Available at:

<https://ifr.org/news/robots-help-to-solve-japans-2024-problem/> [Accessed 4 June 2026].

Nxumalo, S. 2022. Emerging African Innovation Leaders' Exchange and Training Programme. Available at: <https://www.universityworldnews.com/post.php?story=20220410212006411> [Accessed 4 June 2026].

Statistics South Africa. 2025. General Household Survey, 2025. Available at:

<https://www.statssa.gov.za/publications/P0318/P03182025.pdf> [Accessed 4 June 2026].

Statistics South Africa. 2026. South Africa's youth and the labour market in Q1 2026. Available at:

<https://www.statssa.gov.za/?p=19526> [Accessed 4 June 2026].

UNESCO. 2024. What you need to know about higher education in Africa. Available at:

<https://www.unesco.org/en/articles/what-you-need-know-about-higher-education-africa> [Accessed 4 June 2026].

Van Embden, E. 2018. Growing the sharing economy. Available at:

<https://www.bizcommunity.com/Article/196/706/183302.html> [Accessed 4 June 2026].

Via Afrika. 2020. Getting to teaching in the 4IR. Available at:

<https://creators.spotify.com/pod/show/viaafrika/episodes/Getting-to-teaching-in-the-4IR-eb6kcm>
[Accessed 4 June 2026].

World Economic Forum. 2025. The Future of Jobs Report 2025. Available at:

<https://www.weforum.org/publications/the-future-of-jobs-report-2025/> [Accessed 4 June 2026].

Appendix A: AI Declaration Draft

Adapt this wording to match the official Eduvos AI declaration form. Do not submit this appendix if your LMS requires a separate form only.

AI tool used: OpenAI ChatGPT/Codex.

Purpose of use: The tool was used to help create a planning structure, generate a mock model draft, organise argument flow, suggest Harvard-style citation placement, and support editing for clarity.

Extent of use: The AI output was treated as a draft learning aid and was not accepted blindly. The final submission should be rewritten, checked against the assignment brief, verified against sources, and edited in the student's own voice.

Student responsibility: I take responsibility for the final submitted work, including the accuracy of facts, citations, interpretation, originality and compliance with Eduvos academic integrity requirements.

Rewrite Checklist Before Submission

- Replace all placeholders with your campus, student number and required file name.
- Rewrite each paragraph in your own voice and add your own examples where appropriate.
- Verify every statistic and source before final submission.
- Ensure every in-text citation appears in the reference list.
- Complete and upload the official AI declaration form if required.
- Run a final grammar, formatting and Turnitin/originality review before submission.